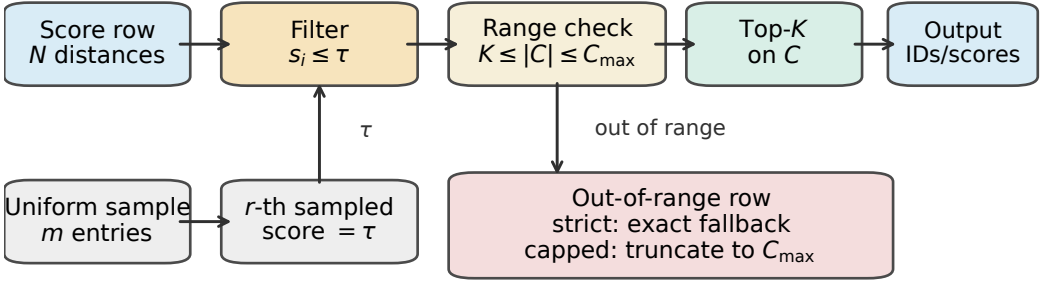
Score row
N distances
Filter
$s_i \le \tau$
Range check
$K \le |C| \le C_{\max}$
Top-K
on C
Output
IDs/scores
τ
out of range
Uniform sample
m entries
r-th sampled
score $= \tau$
Out-of-range row
strict: exact fallback
capped: truncate to $C_{\max}$